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Ayush Roy

AR/VR Research | Immersive Analytics | CSCW | 3DUI

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I am a second-year PhD student in Computer Science at Virginia Tech, advised by Dr. Doug Bowman. My research explores how people organize information, make sense of complex data, and collaborate in immersive environments, with an interest in how AI can further support these activities. I design and evaluate XR interaction techniques through controlled experiments and mixed-methods user studies, with the long-term goal of making immersive computing a practical alternative to traditional desktop computing.

Education

Ph.D., Computer Science, advised by Dr. Doug Bowman — *January 2025 — Spring 2029(expected graduation)*
Virginia Tech

M.Eng., Computer Science — *January 2023 — December 2024*
Virginia Tech *(Transferred to Ph.D. program)*

B.Tech., Information Technology, — *August 2016 — December 2020*
Guru Gobind Singh Indraprastha University (GGSIPU), India

***Relevant Coursework**: Software Engineering, AI Tools for Software Delivery, Big Data Engineering, Information Visualisation, Machine Learning, Agentic AI, Usability Engineering, Advanced topics in XR, Statistics in Research*

Skills

HCI/UX Research	Mixed-Methods Studies, Experimental Design, Qualitative Coding(inductive, deductive, hybrid) Semi-Structured Interviews, Think-Aloud, Survey Design, Figma
XR Prototyping	Unity, OpenXR, C#, VR/AR Interaction, MRTK, Meta XR SDK, Android XR
Analytics	Eye Tracking, Gaze Analysis, Behavioral Log Analysis, Statistical Modeling, Data Visualization, Python, Pandas, ATLAS.ti
AI & ML	PyTorch, TensorFlow, Scikit-learn, Multi-Agent Systems, LLM Integration, RAG, LangChain, LangGraph

Publications

Towards AI-Assisted Organization in Immersive Computational Notebooks — Present
(Under Review)

- Designed and conducted a **user study**(N=24) evaluating **3 levels of AI involvement** (Manual, Suggestion, Predefined) in organizing computational notebook cells, including assisting with **semi-structured interviews** to explore participants' experiences and strategies.
- Contributed to developing an AI recommendation pipeline that identifies semantically related notebook cells and suggests **4 spatial organization structures** (Single Linear, Multiple Linear, Branch, Hierarchy).

Understanding Asynchronous Collaborative Sensemaking Through a Multi-Session Exploratory Study in Immersive Workspaces — Present
(In Preparation for CSCW 27)

- Developed an XR system for asynchronous collaborative data exploration with data logging pipelines for user behavior analysis.
- Conducted a multi-session mixed-methods exploratory study(N=30), combining quantitative behavioral analysis with semi-structured interviews and qualitative analysis to investigate **sensemaking strategies, challenges with handoffs, and collaboration patterns** as analysts built on each other's work.

Characterizing Cognitive Load in Immersive Spaces to Think — Present
(under review)

- Contributed to study design and data analysis strategies for the study(N=24), investigating cognitive load through eye tracking, physiological signals, interaction behaviors, and subjective workload measures.

Evaluating Link Visualization Techniques for Sensemaking in Immersive Workspaces — March 2026
(under review)

- Designed and conducted a controlled 2×3 user study (N=35) evaluating **link routing and bundling** on graph tidiness and comprehension in immersive workspaces.
- Found that **many-to-many edge bundling and non linear link routing** provides the most consistent benefits across tidiness, workload, usability, and completion time without reducing comprehension accuracy.

Exploring Organizational Strategies in Immersive Computational Notebooks — Oct. 2025
S. In, A. Roy, E. Krokos, K. Whitley, C. North, Y. Yang
IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2025)

- Co-designed the experimental study and conducted an exploratory user study on spatial organization in immersive computational notebooks, identifying a preference for **half-cylindrical structures** and significantly greater engagement in non-linear analysis.
- Performed initial data analysis, identifying that participants increasingly adopted **multiple concurrent non-linear**

analytical approaches as notebook size increased, informing design guidelines for immersive data science workspaces.

Other Research Projects

Visualizing Collaborative Spatial Negotiation in XR
Present

- Leading the project in preparation for the **IEEE VR 2027 3DUI Contest**.
- Designing **dual-proposal visualizations** and dynamic rubber-band feedback to communicate spatial disagreements, incorporating perspective switching to support negotiation and convergence toward a shared placement.

Multi-Agent Debiasing Framework with Online Search-Based Validation
Fall 2025

Skills: Python, GPT-4, GPT-3.5-Turbo, Claude-3-Haiku, DeepSeek R-1, LLM Agents, DuckDuckGo API, RAG | GitHub

- Designed **MABRF**, a multi-agent framework with a **two-agent core** (Judge + Rephrase) for automated detection and iterative debiasing of LLM chain-of-thought reasoning.
- Integrated a **Search Agent** via DuckDuckGo API to retrieve real-world evidence before each rephrase pass; built a data pipeline across 110 BBQ-Hard instances and SQuAD v2, automating preprocessing, scoring (Exact Match, F1, Bias Score), and result aggregation.
- Evaluated four LLMs: baseline reduced mean bias score from 0.062 to 0.008 (+3.7% accuracy); GPT-4 + search agent raised EM accuracy from 27% to 45%.

Understanding How Subtle Gaze Direction Cues Shapes Attention and Interpretation in VR
Spring 2026

- Designed a **within-subject VR eye-tracking study(N=18) with 6 cue conditions** (3 cue sizes × 2 luminance levels) to evaluate how subtle visual cues redirect attention without explicit user awareness.
- Analyzed **4 gaze and behavioral measures**: gaze hit rate, time-to-hit, button-confirmation rate, and pre/post gaze-velocity change; identified a **6-degree, 0.625-luminance** cue as the preferred balance between effectiveness and subtlety.
- Designed a follow-up **3-condition 360° narrative study** examining whether covert gaze guidance can influence character perception, trust, narrative understanding, and attentional integrity.

Industry Experience

Software Engineer, Infosys, *Pune, India* *March 2021 — January 2023*

- Automated financial computations via a data pipeline for spreadsheet uploads and real-time processing, reducing computation time from 60s to 200ms for **400,000+ records**; achieved **20% improvement** in efficiency via REST API integration. **Tech: ASP.NET MVC, Angular.js, T-SQL, Azure DevOps.**
- Recognized with the **Pfizer Champ Award** and **Rise Insta Award** in September 2021.

Leadership and Service

President, Student Council *January 2026 — Present*

Center for Human-Computer Interaction at Virginia Tech (CHCI@VT), Blacksburg, VA

- Led council operations spanning faculty/student communication, event planning, and community building.
- **Hack & Yak:** Organized informal gatherings to foster collaboration between HCI students and faculty.
- **Cross-Departmental Workshops:** Collaborated with VT Libraries and SAIG to deliver workshops on AI-assisted research, R, and JMP statistical analysis.
- **Conference Practice Talks:** Facilitated presentation prep sessions for students submitting to venues such as CHI.
- **CHCI Research Symposium (April 2025):** Organized a symposium with **22 presentations** and **75 attendees**, providing a platform for peer and faculty feedback on student research.

Judge *March 2026*

AWS×Virginia Tech CS Careers Hackathon: Evaluated student projects; provided feedback across competing teams.

Reviewer

IEEE International Symposium on Virtual Reality and 3D User Interfaces(VR) 2025, 2026
IEEE International Symposium on Mixed and Augmented Reality (ISMAR) 2025, 2026
ACM Conference on Intelligent User Interfaces (ACM IUI) 2025
International Conference on Extended Reality (ICXR) 2026